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COMPUTER NETWORKS

Ex.No.4	**a) Setup and configure a LAN (Local area network) using a Switch and Ethernet cables in your lab.** 1. Connect 3-4hostmachines toaswitch. 2. Assign ip addresses to each host machine. 3. Check the connectivity between the machines by using ping command. 4. Share and access files and folder across the machines of the LAN.
Date:	

LAN:
Alocal area network (LAN) is a collection of devices connected together in one physical location,

such as a building, office, or home. A LAN can be small or large, ranging from a home network with one user to an enterprise network with thousands of users and devices in an office or school.

Regardless of size, a LAN's single defining characteristic is that it connects devices that are in a single, limited area.

Set Up a LAN (Local Area Network) Directly
HowtoSetUpaLAN(LocalAreaNetwork)

Step 1: Plan the Network
Planand designan appropriate network topology based on the network requirements and the physical location of the devices.
Step 2: Gather the Required Devices
Prepare thefollowing hardwarecomponents:

- 4 Computers (PCs)
- 1 Network Switch (8-port,16-port, or 24-port)
- 4 Ethernet (UTP) Cables

Step 3: Connect the Devices
Connect each computer to thenetwork switch using Ethernet cables. Plug one end of the

Ethernet cable into the Ethernetport of the computer and the other end into an available port on the switch. Ensure that the linkLEDs on the switch and the computers are ON, indicating successful connections.

Step 4: Assign IP Addresses to the PCs

1. Logintoeachcomputerusingan **Administrator** or **Owner** account.
2. Open **Control Panel → Network and Internet → Network and Sharing Center**.
3. Click **Change Adapter Settings**.
4. Right-click **Local Area Connection** (or **Ethernet**)andselect **Properties**.
5. Select **Internet Protocol Version 4 (TCP/IPv4)** andclick **Properties**.
6. Select the **Use the following IP address** option.
7. Enter the IP address and subnet mask for each computer as shown below:

Computer	IP Address	Subnet Mask
PC1	10.1.1.1	255.0.0.0
PC2	10.1.1.2	255.0.0.0
PC3	10.1.1.3	255.0.0.0
PC4	10.1.1.4	255.0.0.0

Click **OK** to save the configuration. Repeat the same procedure for all the PCs connected to the switch.

Step 5: Configure the Network Switch

1. Connect aco mputer tothe switchusing an Ethernet cable.
2. Open a web browser and enter theIP address of the switch in the address bar to access the switch's management interface.
3. Log in using the appropriate **username** and **password**.
4. Configure the basic settings of the switch.
5. Assign the following management IP address to the switch:
6.

Device	IP Address	Subnet Mask
Switch	10.1.1.5	255.0.0.0

Save the configuration.

Step 6: Verify Network Connectivity

Open the **Command Prompt** on eachcomputer and use the **ping** command to test connectivity with the othercomputersandthe switch.

Example:

- `ping`
- `10.1.1.2`
- `ping`
- `10.1.1.3`

If replies are received without packet loss, the network has been configured successfully.

Step 7: Share a Folder

1. Create or select a folder to be shared.
2. Right-click the folder and select **Properties**.
3. Open the **Sharing** tab.
4. Click **Share**.
5. Select **Everyone** fromthe list of users.
6. Click **Share** and then**Done**.

The selected folder is nowaccessible to all computers on the same LAN.

Step 8: Access the SharedFolder

1. Open **File Explorer**onanother computer connected to the LAN.
2. In the address bar,typethe IP address or computer name of the shared computer. For example:
   - `\\10.1.1.1`
   - `\\PC1`
3. Press **Enter**toview the shared folder.
4. Open the folderand verify that files can be accessed successfully.

Procedure to Set Up a LAN Using Cisco Packet Tracer

Step 1: Design the Network Topology

1. Open **Cisco PacketTracer** .
2. From the **NetworkDevices** section, select **Switches** and drag one **2960 Switch** onto the workspace.
3. From the **EndDevices** section, drag four **PCs** onto the workspace.
4. Rename the devices as **PC0,PC1,PC2,andPC3** .

Step 2: Connect the Devices

1. Click (Connections icon).
2. Select . **CopperStraight-ThroughCable**
3. Connect the devices as follows:

Source Device	Interface	Destination Device	Interface
PC0	FastEthernet0	Switch0	FastEthernet0/1
PC1	FastEthernet0	Switch0	FastEthernet0/2
PC2	FastEthernet0	Switch0	FastEthernet0/3
PC3	FastEthernet0	Switch0	FastEthernet0/4

4. Waituntil thelinkLEDsturn **green**, indicatingthatthedevices are successfully connected.

Step 3: Assign IP Addresses to the PCs

ForeachPC:

1. Click thePC.
2. Select the**Desktop**tab.
3. Click **IPConfiguration**.
4. Enter thefollowingIPaddresses:

Device	IP Address	Subnet Mask
PC0	10.1.1.1	255.0.0.0
PC1	10.1.1.2	255.0.0.0
PC2	10.1.1.3	255.0.0.0
PC3	10.1.1.4	255.0.0.0

Leave the **Default Gateway** blankbecause no router is used in this LAN.

Step4:VerifyIP Configuration

OneachPC:

1. Open **Desktop**.
2. Select **Command Prompt**.
3. Execute the following command:

Verify that the configured IP addressand subnet mask are displayed correctly.

Step 5: Test Network Connectivity
From **PC0**,open **CommandPrompt** and execute:

```
ping 10.1.1.2
```
Similarly, test connectivity with the remaining PCs:
```
ping 10.1.1.3

ping 10.1.1.4
```
Repeat the ping test from each PC to ensure that all hosts can communicate successfully.

Step 6: Verify Packet Transmission Using Simulation Mode

1. Click **Simulation** mode at the bottom-right corner of Packet Tracer.
2. Select the **Add Simple PDU** tool (Envelope icon).
3. Click **PC0** as the source device.
4. Click **PC3** as the destination device.
5. Click **Auto Capture/Play**.

Observe the ICMP Echo Request and Echo Reply packets travelling through the switch.

Step 7: Configure an FTP Server for File Sharing
Since Packet Tracer does not support Windows foldersharing, use an FTP server to simulate file sharing.

1. Drag one **Server** from **End Devices** into the workspace.
2. Connect the server to the switch using a **Copper Straight-Through Cable**.
3. Configure the server with:

Device	IP Address	Subnet Mask
Server0	10.1.1.5	255.0.0.0

4. Open **Server0**.
5. Select **Services → FTP**.
6. Turn the FTP service **ON**.
7. Create a username and password (for example, **student / student123**).

Step 8:AccesstheFTPServer
On **PC0**:

1. Open **Desktop → CommandPrompt**.
2. Connect to the FTP server using:

```
ftp 10.1.1.5
```

3. Enter the configured username and password.
4. Execute FTP commands such as:

```
dir
get sample.txt
put report.txt
```

Verify that files can be uploaded to and downloaded from the FTP server.

Step9:Savethe Project

1. Click **File → SaveAs** .
2. Save the Packet Tracer file with an appropriate name, for example:

```
LAN_Configuration.pkt
```

The LAN configuration is now complete.

RESULT: The LAN has been configured using switch and an ethernet. The study of the LAN and switch was also studied.

